Artificial Intelligence 2: Bayesian Networks – Approximate Inference

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Outline of Topics

- Please note that, this lecture is an extra lecture, which will not be part of the exam.
- In Week 9 Lecture 1, we learned how to infer complex relationship in Bayesian networks using exact inference methods.
- As we learned in Week 9 Lecture 1, the complexity of the exact inference grows exponentially with the number of non-evidence random variables. Although there are a few improved exact inference methods, they still have high complexity and sometimes are not tractable.
- In this lecture, we will introduce another way of inference in Bayesian Networks, that is approximation. In particular, we will learn how to use Monte Carlo Sampling for Bayesian Network inference. We will introduce two methods, prior sampling and rejection sampling.

Mathematical Notations

We are using the following shorthand mathematical notations for joint probability distributions of two random variables X and Y using notations of probability:

- The marginal PMF of X: $P(X)$
- The marginal probability of X takes the value x: $p(x)$.
- Joint PMF: $P(X, Y)$
- The value of joint PMF $p(X, Y)$ at (x, y), i.e., $P_{XY}(X = x, Y = y)$: $p(x, y)$
- Conditional PMF: $P(X|Y)$
- The value of conditional PMF $p(X|Y)$ at (x, y), i.e., $P_{X|Y}(X = x, Y = y)$: $p(x|y)$

Approximate Inference by Monte Carlo simulation

Idea: replace the marginalisation (i.e., the summation over non evidence variable assignments) with a **random sample** [1], which is used to approximate the probability.

- **Problem**: Sampling random variables governed by complicated joint probability functions defined by Bayesian Networks .
- **Solution – Monte Carlo Simulation**: "*a computational technique based on constructing a random process for a problem and carrying out a* **NUMERICAL EXPERIMENT** *by N-fold sampling from a random sequence of numbers with a* **PRESCRIBED** *probability distribution*" [2]

[1] In statistics, a random sample is a subset of observations chosen from a larger set. One random sample consists of K elements or observations from the population chosen in such a way that every set of K observations has an equal chance of being the sample actually selected (See Applied Statistics). Sometimes people use "samples" instead of "elements" or "observations".

[2] Jasmina L. Vujic Monte Carlo Sampling Methods

- We we have discussed in our previous lecture, the exact inference of Bayesian Networks by enumeration is a NP hard problem, which means the calculation will grow exponentially with the number of non-evidence variables.
- We mentioned several improved exact inference methods in Probabilistic Graphical Models : Variable elimination, Clustering algorithm and Sum-product algorithm.
- If you have read those chapters, you know these methods still have a high complexity and sometimes are intractable.
- One solution is to approximate the probability we want to infer by random simulation, or more precisely, using Monte Carlo simulation. In science and engineering, especially in physics, if a problem has a PROBABILISTIC interpretation, then we can model this problem using RANDOM NUMBERS. This is the idea of Monte Carlo simulation.
- The idea is to replace the marginalisation, the most time consuming part of the inference, that is, the summation over non evidence variable assignments) with a **random sample**
- Now the problem is how to obtain this random sample. It is not easy since we need to sample random variables governed by complicated joint probability functions defined by Bayesian Networks.
- The solution is Monte Carlo simulation, which is "*a computational technique based on constructing a random process for a problem and carrying out a* **NUMERICAL EXPERIMENT** *by N-fold sampling from a random sequence of numbers with a* **PRESCRIBED** *probability distribution*"

Monte Carlo simulation

Essence of Monte Carlo Simulation: computational simulations that randomly sample a sample space Ω with infinite number of possible outcomes:

- A simple example: rolling 10 dice and what is the probability that the sum is 32?
  - You can calculate it manually (how?); but
  - A lazy solution (Monte Carlo Simulation) : simulate the rolling of 10 dice 10,000 times and count how many of times the 10 dice totalled 32 and then divide it by 10,000.
- Tomorrow's stock market index (FTSE 100) closed price Python Tutorial

Aim: to obtain a unbiased representative subset of the sample space

Please watch this video: Monte Carlo Simulation

- The above definition consists some jargon which need to be explained. However, instead of going to define and explain those jargon, I would like to give you some examples about Monte Carlo Simulation.
- The first example is very simple. Suppose you have an math or probability exam question, rolling 10 dice and calculate the probability that the sum is 32. You can use what you learned in probability to solve this question, but if you happen have 10 dice with you, you can role these 10 dice 10,000 times and count how many of times the 10 dice totalled 32 and then divide it by 10,000.
- You certain can solve the problem without knowing anything about probability but you might run of your exam time.
- The second practical example is to simulate or even predict stock market index, for example FTSE100 closed price. The Monte Carlo simulation works by simulating random walks or Brownian Motions. To simulate Random walk of the stock market, we also need to know the daily growth rate and standard deviation, which can be calculated from the historical data. We then feed these two parameters into a random walks model and generate random value of tomorrow's FTSE100 closed price based on today's closed price.
- I will not go into the details of this but you can check out the tutorial listed here.
- From the above two examples, we shall learn that the aim of Monte Carlo simulations is to obtain a unbiased representative subset of the sample space. The sample space could be complex but we will have some techniques to simplify the the sampling.
- You can watch this tutorial on Monte Carlo Simulation here.

Monte Carlo Sampling

Monte Carlo Sampling: We sample from a joint distribution using the following two steps:

- Step 1: Generate a random sequence of N numbers with a prescribed probability distribution, e.g., continuous uniform distribution on $[0, 1)$
- Step 2: Generate N observations from requested sample distribution

Question: given a distribution of Cloudy as below, which is a Bernoulli distribution $P(C) \sim \text{Bernoulli}(0.6)$, how to use the Monte Carlo Sampling method to generate a random sample from this distribution?

$P(C = 0)$	$P(C = 1)$
0.4	0.6

- To conduct Monte Carlo Sampling, we use the following two steps to sample from a complicated probability distribution.
- The first step is to generate a sequence of random numbers with a prescribed probability distribution. We usually use a very simple probability distribution, such as the continuous uniform distribution on $[0, 1)$.
- We then use the sequence of random numbers to generate N observations from the requested, and usually complicated distribution.
- Here is an example, which the requested sample distribution is not complicated at all, but we use this simple example to illustrate the idea of Monte Carlo Sampling.
- The example is from the Wet Grass Bayesian Network example we have already seen. Here, we are interested in generating the probability distribution of cloudy, which as we know is essentially a Bernoulli distribution $P(C) \sim \text{Bernoulli}(0.6)$ as we learned in Lecture 2. So how to use the Monte Carlo Sampling method to generate a random sample from this distribution?

Sampling from a joint distribution

Solution: we follow the above two steps:

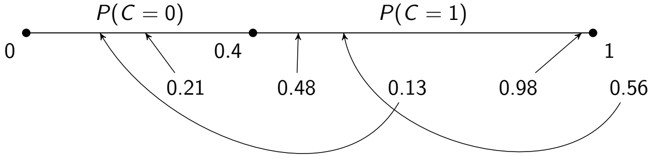
- Step 1: Generate a random sequence of $N = 5$ numbers by drawing from a continuous uniform distribution on $[0, 1)$

0.21	0.48	0.13	0.97	0.56

- Step 2: Generate observations from the requested sample distribution $P(C) \sim \text{Bernoulli}(0.6)$

Random Number	0.21	0.48	0.13	0.97	0.56
C	0	1	0	1	1

$P(C=0)$	$P(C=1)$
0.4	0.6



- To generate a random sample, we follow the above two steps of Monte Carlo Sampling. Let's say we need 5 observations, we first generate a sequence of $N = 5$ random numbers by drawing from a continuous uniform distribution on $[0, 1)$, which are tabulated in the following table.
- The second step is to generate observations from the requested sample distribution $P(C) \sim$ Bernoulli(0.6), which can be done by a simple if-then-else line of code to implement. In this example, we know that $P(C = 0) = 0.4$, which means we can map those random numbers that are smaller than 0.4 to the event that it is cloudy, that is $C = 0$, otherwise, $C = 1$.
- This can be illustrated by the following graph, which maps the continuous uniform distribution on $[0, 1)$ to the Bernoulli distribution Bernoulli(0.6)
- As you can see, the frequency of observations of $C = 1$ is 0.6, which follow the Bernoulli distribution Bernoulli(0.6).

BN Approximate Inference by Prior Sampling

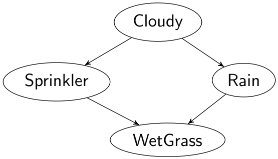
To draw a random sample with N observations from the joint distribution $P(X_1, X_2, \ldots, X_n)$, if we do not have any evidence, i.e., $\boldsymbol{e} = \emptyset$, we have the following algorithm:

Algorithm 1: Prior Sampling (Sampling without evidence)

inputs: a BN specifying joint distribution $P(X_1, X_2 \ldots, X_n)$
for $j = 1$ **to** N **do**
  for $i = 1$ **to** n **do**
    $x_{ji} \leftarrow$ Use Monte Carlo Sampling method to generate value from $P(X_i|\text{Parents}(X_i))$
Return a random sample with N observations, each is defined as $\boldsymbol{x}_j = [x_{j1}, x_{j2}, \cdots, x_{jn}]$, $j = 1, \ldots N$

- Now, let's move on to study how to sample a complicated joint distribution represented by a Bayesian Network.
- Here we assume the Bayesian network has no evidence, that is, we don't have the values of the evidence variable $\boldsymbol{E}$, which also means the Bayesian network is empty. This is why we call this algorithm Prior Sampling, or sampling without evidence.
- Assuming we want to draw a random sample of N observations, here is the pseudo-code of this algorithm.
- The input is the joint distribution represented by a Bayesian Network, and in the algorithm, we have a double loop where the outer for loop generates N observations. The inner for loop use the Monte Carlo Sampling method we just learned to generate value for each variable based on the local conditional probability distributions, that is $P(X_i|\text{Parents}(X_i))$
- After this double-loop finish running, we will have a random sample with N observation and each observation, i.e., $\boldsymbol{x}_j$ is instantiation of the joint probability distribution.
- Let's see an example to help you understand this algorithm.

Example: Wet Grass Bayesian Network



We first need to determine the order of the nodes to map to the sequence of random numbers, which could be

- $\{C, R, S, W\}$
- $\{C, S, R, W\}$

We choose $\{C, R, S, W\}$. In total: $n = 4$ nodes, we generate a sequence of 4 random number from $[0, 1)$

0.21	0.84	0.07	0.98
C	R	S	W

- Let's use the Wet Grass Bayesian network as our example to show you how to do prior sampling.
- The first decision is how to decide the order of those nodes in the network so we can map the sequence of the random numbers to the joint probability.
- For this simple network, we only have two choices:
  - $\{C, R, S, W\}$
  - $\{C, S, R, W\}$

  and we select the first one.
- Since there are 4 nodes in total, we need to generate a sequence of 4 random numbers uniformly sample from the $[0, 1)$
- Here is an table that map those random numbers to the nodes. We also use colours to help us better visualise it.

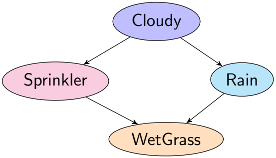
Example

Execute the inner loop of Algorithm 1, we have

0.21	0.84	0.07	0.98
$C = 0$	$R = 1$	$S = 0$	$W = 1$

$P(C=0)$	$P(C=1)$
0.4	0.6

C	$P(S=0)$	$P(S=1)$
0	0.5	0.5
1	0.9	0.1



C	$P(R=0)$	$P(R=1)$
0	0.8	0.2
1	0.2	0.8

S	R	$P(W=0)$	$P(W=1)$
0	0	1.0	0.0
1	0	0.1	0.9
0	1	0.1	0.9
1	1	0.01	0.99

- We then follow the pseudo-code of Algorithm 1 on page 8. We let $j = 1$ and we only execute the inner loop to draw one observation $\boldsymbol{x}_1$.
- We start with the first node, Cloudy. Since the random number is 0.21, from what we have seen in the simple example on page 7, we know the value of C should be 0.
- Then the next variable or the second node is Rain. We now have $C = 0$, then we only look at the first row of this CPT. Since the second random number is 0.84, which is larger than 0.8, we know from the Monte Carlo Sampling we learned, it should map to the event $R = 1$.
- Then the third variable or node is Sprinkler. Again, we only use the first row with $C = 0$, and look at the third random number, which is 0.07, we know it should map to the event $S = 0$.
- Finally, we move to the last node, WetGrass. We now have $S = 0$ and $R = 1$, so, we look at the 3rd row in the CPT. Since the 4th random number is 0.98, we know $W = 1$
- Now we have completed the inner loop of Algorithm 1 on page 8 and generated one observation $\boldsymbol{x}_1$.

Example

Execute the outer loop of Algorithm 1 with $N = 10$, we obtained the following random sample:

$C = 0$	$R = 1$	$S = 0$	$W = 1$
$C = 0$	$R = 0$	$S = 1$	$W = 1$
$C = 1$	$R = 1$	$S = 0$	$W = 1$
$C = 1$	$R = 0$	$S = 0$	$W = 0$
$C = 1$	$R = 1$	$S = 0$	$W = 1$
$C = 0$	$R = 0$	$S = 1$	$W = 1$
$C = 1$	$R = 1$	$S = 1$	$W = 1$
$C = 1$	$R = 1$	$S = 0$	$W = 1$
$C = 0$	$R = 0$	$S = 0$	$W = 0$
$C = 1$	$R = 1$	$S = 0$	$W = 1$

We can infer by counting the number of observations. For example, to infer the probability of grass is wet $P(W = 1)$, we count the number of observations that $W = 1$ and calculate the probability as

$$P(W = 1) \approx \frac{8}{N} = \frac{8}{10} = 0.8$$

- We continue the outer loop from 1 to N, and say $N = 10$, we can generate a random sample with 10 observations.
- Now, the inference can be done by a primary school student who know how to count.
- Let's try a simple example, to infer the probability of grass is wet $P(W = 1)$, we simply count how many observations that $W = 1$ and then divided by the total number of observations, that is $N = 10$.
- By simple counting, we know there are 8 observations where $W = 1$ and the probability or more precisely, the frequency of $W = 1$ is 0.8.

Example

Given the following random sample:

$C=0$	$R=1$	$S=0$	$W=1$
$C=0$	$R=0$	$S=1$	$W=1$
$C=1$	$R=1$	$S=0$	$W=1$
$C=1$	$R=0$	$S=0$	$W=0$
$C=1$	$R=1$	$S=0$	$W=1$
$C=0$	$R=0$	$S=1$	$W=1$
$C=1$	$R=1$	$S=1$	$W=1$
$C=1$	$R=1$	$S=0$	$W=1$
$C=0$	$R=0$	$S=0$	$W=0$
$C=1$	$R=1$	$S=0$	$W=1$

Question: Infer the following probability:

$$P(R=1, W=1)$$

$$P(R=1|W=1)$$

$$P(W=1|C=1)$$

- Here are a few more complex example. Please pause this video for 5 mins and try to complete yourself.

Example

Execute the outer loop of Algorithm 1 with $N = 10$, we obtained the following random sample:

$C=0$	$R=1$	$S=0$	$W=1$
$C=0$	$R=0$	$S=1$	$W=1$
$C=1$	$R=1$	$S=0$	$W=1$
$C=1$	$R=0$	$S=0$	$W=0$
$C=1$	$R=1$	$S=0$	$W=1$
$C=0$	$R=0$	$S=1$	$W=1$
$C=1$	$R=1$	$S=1$	$W=1$
$C=1$	$R=1$	$S=0$	$W=1$
$C=0$	$R=0$	$S=0$	$W=0$
$C=1$	$R=1$	$S=0$	$W=1$

Answer:

$$P(R=1, W=1) \approx \frac{6}{10} = 0.6$$

$$P(R=1|W=1) \approx \frac{P(R=1, W=1)}{P(W=1)} = \frac{6}{8} = 0.75$$

$$P(W=1|C=1) \approx \frac{P(C=1, W=1)}{P(C=1)} = \frac{5}{6} \approx 0.8$$

- Again, to solve these inference questions, we just need to count. For the first question, we simply count how many observations that have $R = 1$ AND $W = 1$, and we have 6 observations, so the approximated probability is $\frac{6}{10} = 0.6$
- For the second question, that is given the grass is wet, what is the probability it rained? You might notice, we solved the second question using Bayes theorem in Week 9 Lecture 2, from page 13 to 16. From that computation, we got the probability 0.75. Here, let me show you how to use this random sample to get a very accurate approximation.
- To approximate this conditional probability, we use the equation of the joint probability of $R = 1$ AND $W = 1$ divided by the probability of $W = 1$, which give use $\frac{6}{8} = 0.75$
- Now the third question is also what we solved before, that is in Week 9 Lecture 1, page 12, where we used exact inference by enumeration. The exact probability is 0.7452.
- Here, we can approximate the probability by simply counting how many observations have $C = 1$ and $W = 1$ and how many observations have $C = 1$. Then we can approximate the conditional probability the grass is wet given it is cloudy as 0.8, a little larger than the exact probability 0.7452.

Problems with prior sampling

As we have seen in page 13, we have the evidence that it is cloudy (i.e., $C = 1$), we want to estimate $P(W = 1|C = 1)$ only, is prior sampling an efficient method?

A problematic solution: the prior sampling algorithm is not efficient because:

- The random sample might not consists of observations with $C = 1$, i.e., number of observations with $C = 1$ is 0 $\Rightarrow P(W = 1|C = 1) \approx \frac{P(W=1,C=1)}{P(C=1)} = \frac{0}{0}$
- Many observations in the random sample are not consistent with the evidence – Waste of computation resource. Example: we draw a random sample with 100 observations, only 63 observations have $C = 1$. The other 37 observations of $C = 0$ will not be used in our estimation

Question: If we know the values of the evidence variables in our query in advance, i.e., we know $P(X|E = e)$, how to sample more efficiently?

Answer: Rejection sampling

- In the previous examples, when we sample, we assume we do not have any values of the evidence variables. However, as we have seen in the example on page 13, our inference or query could be very specific, that is, we have some value of the evidence variables, and we ask for the conditional probability. So, if have an inference problem that we know the values of the evidence variables in the query, is the prior sampling an efficient method?
- The answer is no. The prior sampling is not an efficient method due to two reasons:
  - The random sample might not consists of observations with $C = 1$, i.e., number of observations with $C = 1$ is 0 $\Rightarrow P(W = 1|C = 1) \approx \frac{P(W=1,C=1)}{P(C=1)} = \frac{0}{0}$ and this fraction is meaningless.
  - We will waste a lot of computational resource, CPU time and memory on those observations that are not consistent with the evidence. For example: we draw a random sample with 100 observations, only 63 observations have $C = 1$. The other 37 observations of $C = 0$ will not be used in our estimation.
- Now the question is, how to sample more efficiently if we know the values of the evidence variables in advance in our query or the inference problems.

Rejection sampling

Idea: We simply reject those observations once an evidence variable has been sampled to take on a value inconsistent with the evidence of the query.

Algorithm 2: Rjection Sampling

inputs: a BN specifying joint distribution $P(X_1, X_2 \ldots, X_n)$
for $j = 1$ **to** N **do**
  for $i = 1$ **to** n **do**
    $x_{ji} \leftarrow$ Use Monte Carlo Sampling method to generate value from $P(X_i|\text{Parents}(X_i))$
    if x_{ji} is not consistent with evidence **then**
      Reject: $j = j - 1$ and **break**
Return a random sample with N observations, each is defined as $\boldsymbol{x}_j = [x_{j1}, x_{j2}, \cdots, x_{jn}]$, $j = 1, \ldots N$

- The idea of reject sampling is very simple and as its name suggested, we simply reject those observations whose values of the evidence variables are not consistent with those in the query, once an evidence variable has been sampled.
- In terms of the pseudo-code, we just need to have an conditional statement to check if the sampled values of the evidence variables are consistent with those in the query or not, if not, then break the inner loop and sample again. Just note that, we still need N observations, so we let the counter j remain unchanged before breaking the inner loop.

Rejection sampling

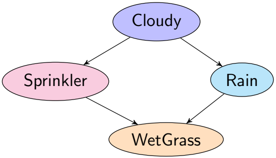
Example: $P(W = 0 | S = 1)$

0.21	0.84	0.07	0.98	Reject/Accept
$C = 0$	$R = 1$	$S = 0$		Reject

$P(C = 0)$	$P(C = 1)$
0.4	0.6

C	$P(S = 0)$	$P(S = 1)$
0	0.5	0.5
1	0.9	0.1

Reject



C	$P(R = 0)$	$P(R = 1)$
0	0.8	0.2
1	0.2	0.8

- Let's take a look at an example of rejection sampling. The inference problem or the query is $P(W = 0|S = 1)$, that is, we know the evidence variable S takes the value of 1 in advance. The query is to obtain the conditional probability of $W = 0$, given $S = 1$.
- This means, we need to reject those observation once the evidence variable S has been sampled to take value $S = 0$, which is inconsistent with the query $S = 1$.
- Similar to what we did using the prior sampling method, we first decide the order of those nodes, that is C-R-S-W, and then generate four random numbers, then we use the local conditional probability distributions $P(X_i|\,\text{Parents}(X_i))$ and the CPTs to map those random numbers to the values of the nodes or random variables.
- We start with the Cloudy node, we map 0.21 to $C = 0$, and we then map 0.84 to $R = 1$, now for Sprinkler, we map 0.07 to $S = 0$, which is inconsistent with $S = 1$ in our query, we reject this sample, which means we stop or break this loop and sample again.

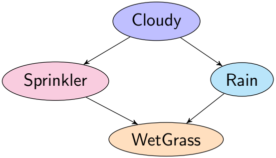
Rejection sampling

Example: $P(W = 0 | S = 1)$

0.56	0.42	0.74	0.29	Reject/Accept
$C = 0$	$R = 1$	$S = 1$	$W = 1$	Accept

$P(C = 0)$	$P(C = 1)$
0.4	0.6

C	$P(S = 0)$	$P(S = 1)$
0	0.5	0.5
1	0.9	0.1



C	$P(R = 0)$	$P(R = 1)$
0	0.8	0.2
1	0.2	0.8

S	R	$P(W = 0)$	$P(W = 1)$
0	0	1.0	0.0
1	0	0.1	0.9
0	1	0.1	0.9
1	1	0.01	0.99

- Since the previous observation is rejected, we repeat the inner loop with a new sequence of 4 random numbers.
- It is similar to what we did, so I will not go into the details of this example.

Problems with rejection sampling

Problems: still expensive when

- $P(e)$ is small $\Rightarrow$ rejects lots of observations!
- Evidence variables have many ancestors $\Rightarrow$ You generated a lot of values x_i from $P(X_i|\text{Parents}(X_i))$ before rejecting a observation.

Solution: Likelihood weighting

- Idea 1: Force the evidence variables to be consistent with the given evidence values
- Idea 2: Re-weight the observation to account for the CPT's of the evidence variables – essentially use the probabilities in the CPTs as the probabilities of observing the values of the evidence variables

Further reading

For approximate inference, please read Probabilistic Graphical Models (find the pdf yourself.)for

- Likelihood weighting: Chapters 12.2
- Markov Monte Carlo Methods: Chapter 12.3

For approximate learning, please read Probabilistic Graphical Models (find the pdf yourself.) for

- Parameter Estimation (maximum likelihood estimation (MLE)):Chapters 17.1 to 17.3
- Structure Learning: 18.1